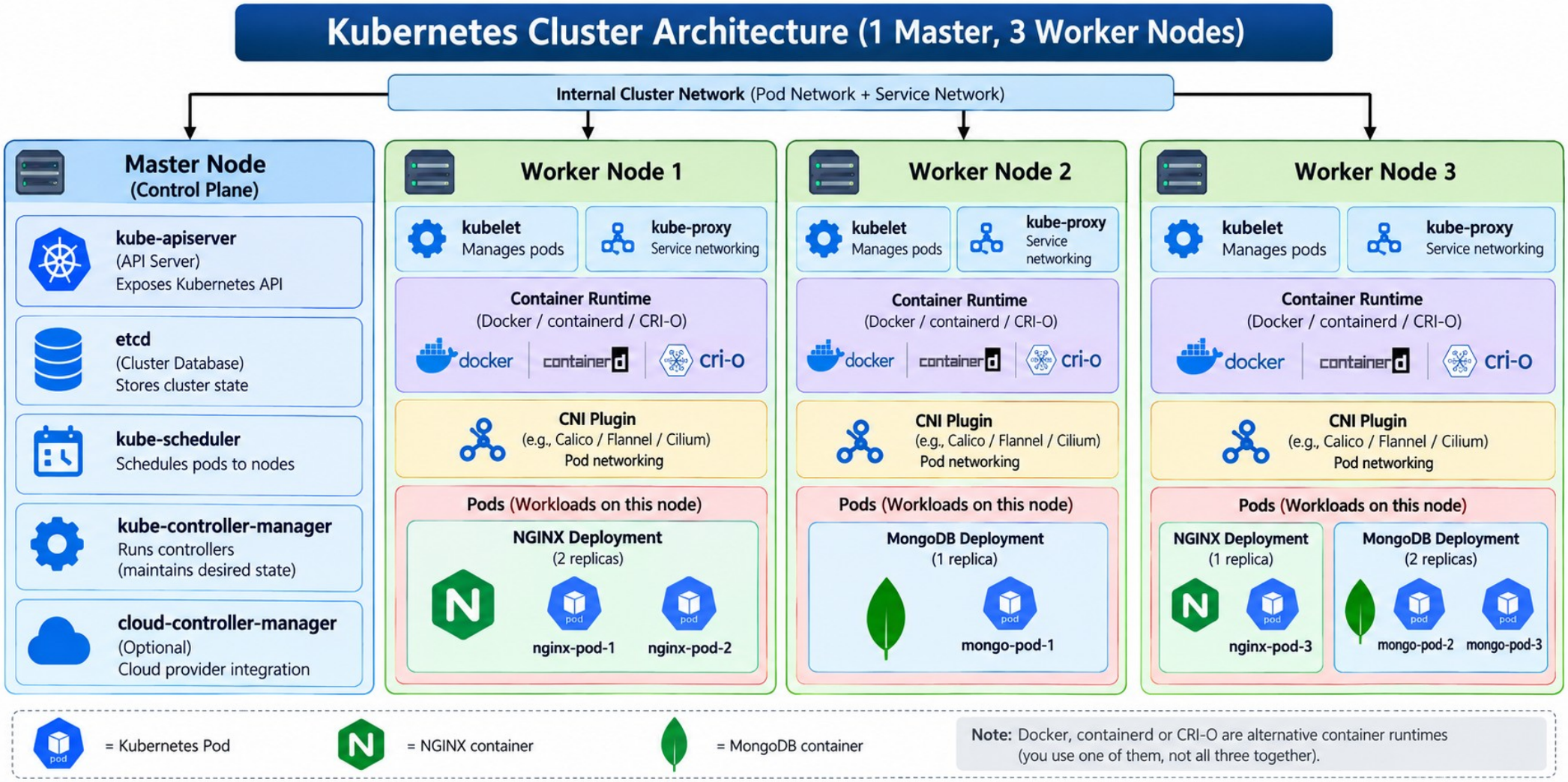

Kubernetes Cluster Architecture (1 Master, 3 Worker Nodes)
Internal Cluster Network (Pod Network + Service Network)
Master Node
(Control Plane)
kube-apiserver
(API Server)
Exposes Kubernetes API
etcd
(Cluster Database)
Stores cluster state
kube-scheduler
Schedules pods to nodes
kube-controller-manager
Runs controllers
(maintains desired state)
cloud-controller-manager
(Optional)
Cloud provider integration
Worker Node 1
kubelet
Manages pods
kube-proxy
Service networking
Container Runtime
(Docker / containerd / CRI-O)
docker
containerd
cri-o
CNI Plugin
(e.g., Calico / Flannel / Cilium)
Pod networking
Pods (Workloads on this node)
NGINX Deployment
(2 replicas)
nginx-pod-1
nginx-pod-2
Worker Node 2
kubelet
Manages pods
kube-proxy
Service networking
Container Runtime
(Docker / containerd / CRI-O)
docker
containerd
cri-o
CNI Plugin
(e.g., Calico / Flannel / Cilium)
Pod networking
Pods (Workloads on this node)
MongoDB Deployment
(1 replica)
mongo-pod-1
Worker Node 3
kubelet
Manages pods
kube-proxy
Service networking
Container Runtime
(Docker / containerd / CRI-O)
docker
containerd
cri-o
CNI Plugin
(e.g., Calico / Flannel / Cilium)
Pod networking
Pods (Workloads on this node)
NGINX Deployment
(1 replica)
nginx-pod-3
MongoDB Deployment
(2 replicas)
mongo-pod-2
mongo-pod-3
= Kubernetes Pod
= NGINX container
= MongoDB container
Note: Docker, containerd or CRI-O are alternative container runtimes
(you use one of them, not all three together).